

VERIFICATION OF THEVENIN'S THEOREM



AIM:

To verify Thevenin's theorem practically and theoretically for the given D.C. circuit.



APPARATUS REQUIRED:

<u>SL. NO.</u>	<u>APPARATUS</u>	<u>SPECIFICATION</u>	<u>QUANTITY</u>
<u>1.</u>	Regulated Power Supply (RPS)	(0-30V)	1
<u>2.</u>	Voltmeter	(0-30V) M.C.	1
<u>3.</u>	Ammeter	(0-10 mA) M.C.	1
<u>4.</u>	Resistors	470 Ω , 560 Ω , 1K Ω	2, 1, 1 ^(respective -ly)
<u>5.</u>	Bread board	—	1
<u>6.</u>	Multimeter	—	1



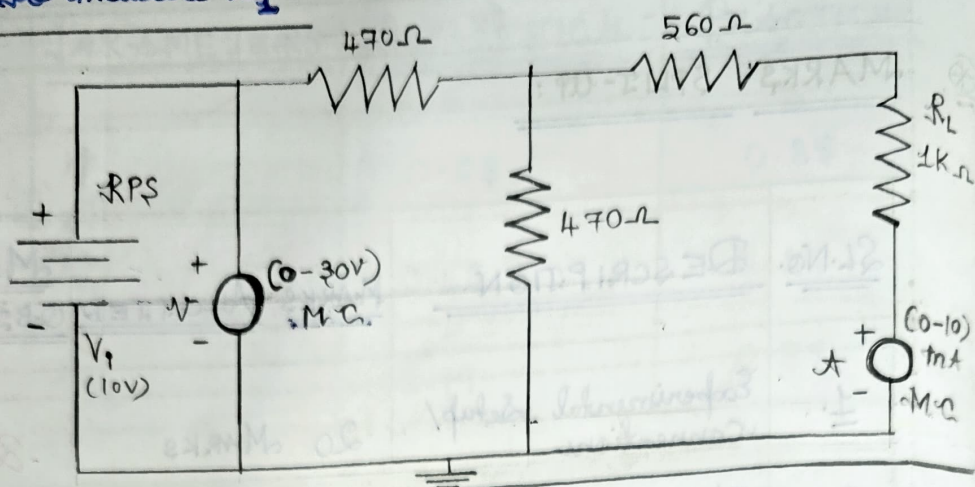
PROCEDURE:

1. Make the connections as per Circuit Diagram: 1.
2. Vary the RPS and set an input voltage of 10V.
3. Note down the voltmeter reading (V_1) and ammeter reading (I_1) in Tabular Column 1.
4. Switch off the supply and make connections for Circuit Diagram 2.
5. Measure the Thevenin's resistance R_{th} = Norton's resistance R_N .
6. Switch off the supply and make connections for Circuit Diagram 3.
7. Set an input voltage of 10V in the RPS and note down the voltmeter readings V_1 and V_{Th} ($=V_{oc}$) in Tabular Column: 3.

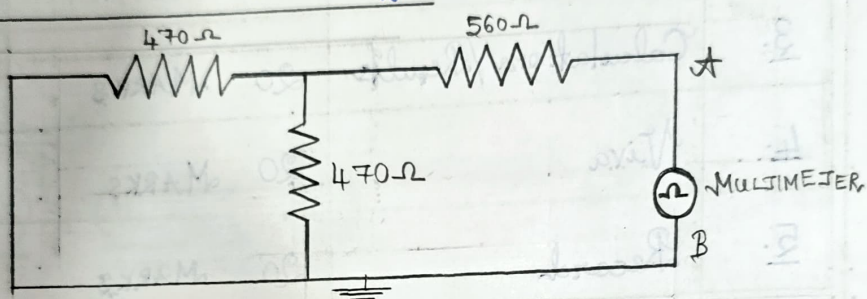


CIRCUIT DIAGRAM:

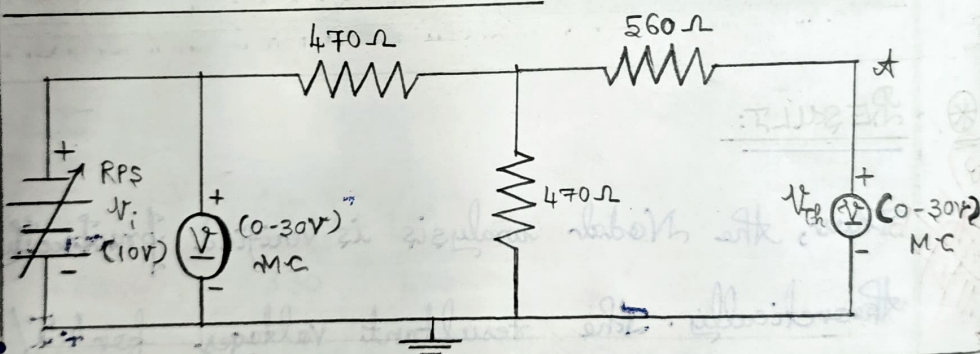
To measure I_L :



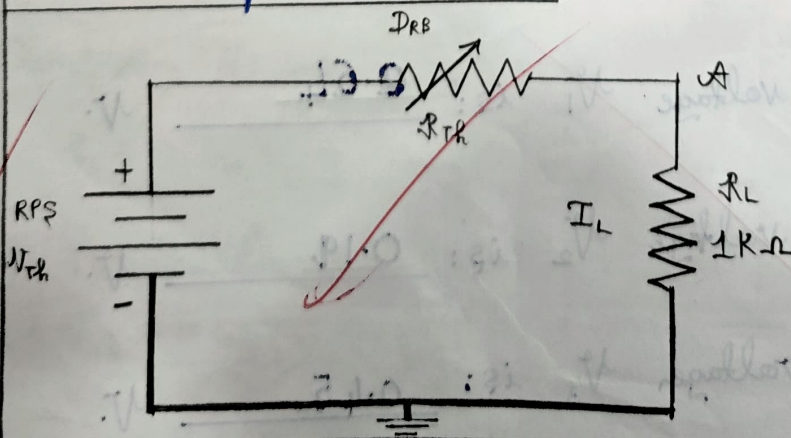
To measure R_{Th} or R_N :



To measure V_{Th} or V_{oc} :



Thevenin's equivalent circuit:



⊛

TABULAR COLUMN 5:

①

To measure I_L :

V_i (Volts)	I_L (Amps)
2 V	0.36 mA
4 V	0.72 mA
6 V	1.08 mA
8 V	1.44 mA
10 V	1.80 mA

②

To measure R_{Th} or R_N :

From circuit diagram 2, $R_{Th} = R_N = \underline{798 \Omega}$.

③

To measure V_{Th} or V_{oc} :

V_i (Volts)	V_{Th} (Volts)
2V	0.65 V
4V	1.30 V
6V	1.95 V
8V	2.60 V
10V	3.25 V

⊛

MODEL CALCULATION:

Practical value of I_L (from tabulation 1) = 2.3 mA.

Verification of Thevenin's theorem:

$$I_L = V_{Th} / (R_{Th} + R_L) = \underline{2.22 \text{ mA}}$$

Theoretical calculation of I_L , R_{Th} (R_N) and V_{Th} for the given circuit:

⇒ For calculating R_{Th} :

$$R_R = \frac{470 \times 470}{470 + 470} = \underline{235 \Omega}$$

$$V_{Th} = \frac{V_i \times R_R}{R_i + R_R} = \underline{3.33 \text{ V}}$$

$$\Rightarrow (\text{Practical}) = \underline{0.325 \text{ V}}$$

⇒ For calculating I_2 :

$$I_2 = \frac{V_{Th}}{R_{Th} + R_L}$$

$$= \frac{3.23}{795 + 1000} = \underline{1.85 \text{ mA}}$$

8. Switch off the supply and make corrections for Circuit Diagram 4.
9. Set an input voltage of 10V in the RPS and note down the voltmeter reading V_i and Ammeter reading $I_N (= I_{sc})$ in Tabular Column 4.
10. Draw the Thevenin's equivalent circuit and Norton's equivalent circuit as shown in circuit diagrams 5 & 6 respectively.
11. Calculate the I_L value using the formula:
 Thevenin's Theorem:

$$I_L = V_{Th} / (R_{Th} + R_L)$$

 Norton's Theorem:

$$I_L = I_N * R_N / (R_N + R_L)$$
12. Theoretically verify the Norton's theorem.

⊗ MARKS SPLIT-UP:

SL. No.	DESCRIPTION	MARKS ALLOTTED	MARKS OBTAINED
1.	Experimental Setup/Connections.	20 MARKS	20
2.	Observation/Readings	20 MARKS	20
3.	Calculations/Results	20 MARKS	20
4.	Viva	20 MARKS	20
5.	Record	20 MARKS	20
		<u>TOTAL:</u>	100

⊗ RESULT:

Thus, Thevenin's theorem is verified practically and theoretically.